

Bhai ab mai tumhe **Cycle Sort Pattern ka MASTER TEMPLATE** de raha hoon. Agar ye samajh liya na, to **10–12 LeetCode questions automatically solve ho jayenge.** 🚀

🧠 Core Idea (Golden Rule)

Agar array me numbers **1 → n range me hain**, to:

value $x \rightarrow$ index $x-1$

Matlab har number ko **uski correct index position par bhejna hai.**

Example:

value : 1 2 3 4 5

index : 0 1 2 3 4

★ Master Cycle Sort Template

```
void cycleSort(vector<int>& nums){
```

```
    int n = nums.size();
```

```
    for(int i = 0; i < n; i++){
```

```
        while(nums[i] > 0 && nums[i] <= n
              && nums[i] != nums[nums[i]-1]){
```

```
            swap(nums[i], nums[nums[i]-1]);
```

```
        }
```

```
    }
```

```
}
```

Meaning:

while element wrong position par hai

swap with correct position

Example

nums = [3,4,-1,1]

Step1

[3,4,-1,1]

swap(3 , -1)

[-1,4,3,1]

Step2

swap(4 , 1)

[-1,1,3,4]

Step3

swap(1 , -1)

[1,-1,3,4]

Now array almost sorted.

First Missing Positive (Cycle Sort)

```
int firstMissingPositive(vector<int>& nums) {
```

```
    int n = nums.size();
```

```
    for(int i = 0; i < n; i++){
```

```
        while(nums[i] > 0 && nums[i] <= n &&
```

```
            nums[i] != nums[nums[i]-1]){
```

```
                swap(nums[i], nums[nums[i]-1]);
```

```
        }
```

```
    }
```

```
    for(int i = 0; i < n; i++){
```

```
        if(nums[i] != i+1)
            return i+1;
    }

    return n+1;
}
```

Complexity

Time : $O(n)$

Space : $O(1)$

Kyuki har element **maximum ek baar correct position par jata hai**.

Cycle Sort Pattern Questions

Ye saare questions **same template se solve hote hain**:

Easy

- 1** Missing Number
- 2** Find All Numbers Disappeared in Array

Medium

- 3** Find All Duplicates in Array
- 4** Set Mismatch

Hard

- 5** First Missing Positive
-

Pattern Recognition Trick (Important)

Agar question me likha ho:

numbers from 1 to n

ya

numbers within range of array size

To **90% chance** hai **Cycle Sort** pattern lagega.

💡 **Golu bhai ek pro tip**

Striver A2Z sheet me **3 patterns** bahut powerful hain:

1 Two Pointer

2 Binary Search on Answer

3 Cycle Sort

Inhe master kar liya to **DSA ka 40–50% medium level** solve ho jata hai.

Agar chaho to mai tumhe **Cycle Sort** ka ek visual diagram bhi bana ke samjha sakta hoon jisse ye pattern **life time** ke liye dimaag me baith jayega. 🧠 ✨